

Philosophical Logic Task C

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Briefly expound and critically assess what you take to be the best argument for:

(a) accepting the Barcan Formula and the Converse Barcan Formula.

Response: I take Williamson's argument in *Bare Possibilia* to be the best argument for accepting the Barcan Formula (BF) and the Converse Barcan Formula. The Barcan Formula is provable in SQML

$$\vdash_{\text{SQML}} \forall x \Box \phi \rightarrow \Box \forall x \phi \quad (\text{BF})$$

By the soundness of SQML, we have that BF is valid in SQML,

$$\models_{\text{SQML}} \forall x \Box \phi \rightarrow \Box \forall x \phi$$

Similarly with the Converse Barcan formula, we have

$$\vdash_{\text{SQML}} \Box \forall x \phi \rightarrow \forall x \Box \phi \quad (\text{BCF})$$

and by soundness,

$$\models_{\text{SQML}} \Box \forall x \phi \rightarrow \forall x \Box \phi$$

Given that BFC and BF are theorems of SQML, they face no possible counterexamples i.e. there is no model $\mathcal{M} = \langle \mathcal{D}, \mathcal{I}, \mathcal{W} \rangle$ in which BF or BCF are false. But represented in English, they seem to be vulnerable to *prima facie* counterexamples. The BF essentially states (in its logically equivalent existential form) that if there could have been an x that is a ϕ , then there is some x that could have been a ϕ . But this seems to be intuitively false. Suppose we take ϕ as 'Wittgenstein fathered x ' and although Wittgenstein did not have children, it is plausible that he possibly could have had a child. But the consequent of BF commits us to picking out an object that could have been fathered by Wittgenstein. And since it seems that your parentage is essential to you as a person, no actual object (person or non-person) that exists in our world could have been fathered by Wittgenstein. The BCF also succumbs to similar counterexamples, though for our purposes I will just focus on BF since BFC can be derived from BF.

Williamson's solution to disarming these purported counterexamples is to commit to the existence of *bare possibilia* or objects that merely exist as possibilities and not as concrete objects. In short, Williamson is committed to the necessity of existence: $\Box \forall x \Box \exists y (x = y)$, and says that in every world w , everything exists. What matters is in what sense they exist – either concretely or as simply *bare possibilia* with no spatiotemporal location. Therefore, the candidate counterexamples to BF and BCF do not hold (Wittgenstein's child exists with no spatiotemporal location), and are instead counterexample to a different claim that if there is something in space and time that could have been ϕ , then there could have been something in space and time that was ϕ .

Such an ontological commitment to the necessity of everything existing, but in different modes seems costly. Furthermore, Williamson must justify this commitment over the alternative to relativise domains to worlds i.e. that $\mathcal{D}(w_1) \neq \mathcal{D}(w_2)$. Here, Williamson's argument falters: he argues that the relativisation of domains to worlds creates great complications for formal semantics and that completeness proofs are "more convoluted". But this is not ipso facto a reason to choose SQML over VDQML, and sacrifice our common-sense intuitions about what exists. Furthermore, Williamson smuggles in a controversial notion of existing

– clearly, what is up for debate is whether bare possibilia do exist in the way that we think $\exists x$ captures. Of course, we do not think that having a spatio-temporal location is necessary to exist (intuitively, numbers exist for instance), but to think that even possibilia without names to refer to them exist seems to do an injustice towards the notion of existence. When I say ‘Matthew exists’, I would not be happy with Matthew existing as possibilia, but an actual Matthew existing in my world. Williamson’s solution has not actually resolved the problem, merely delayed it by making the problem about what counts as existence.

(b) rejecting the Barcan Formula and the Converse Barcan Formula

Response: I take Sider’s argument in Williamson’s many necessary existents to be the strongest argument for rejecting the Barcan Formula and the Converse Barcan Formula. Sider’s argument runs as an argument by reductio.

1. If we accept an iterative conception of a set, there is a set of all nonsets
2. Lewis’ modal realism licences a principle of recombination
3. Modal Realism applies to Williamson’s view about many necessary existents
4. The recombination principle, together with BF and BCF (in its infinitary form), entail that if we attempt to construct a set of all nonsets with cardinality v , we can construct a set with cardinality greater than v that contains only nonsets, and thus contradicting our assumption that there is a set of all nonsets
5. Hence, we should not accept Williamson’s argument for accepting BF and BCF, and should reject them

I am going to challenge premise (3) and the assumptions that allow for premise (4) to hold true. In particular, the tenor of my argument is just because Lewis’s modal realism and the auxillary assumptions Sider makes are consistent with Williamson’s argument, does not mean that Williamson’s argument entails them or requires them.

Let us begin with the assumptions Sider takes Williamson to endorse:

$$\begin{aligned} \Box \forall x (Sx \rightarrow \Box Sx) & \quad \text{(ES)} \\ \Box \forall x \forall y (x = y \rightarrow \Box x = y) & \quad \text{(NI)} \end{aligned}$$

Since by Sider’s admission that these assumptions are not strictly required by Williamson’s argument, we can freely assess whether to reject them. Specifically, I suggest that the principle of the necessity of identity is not at all obvious. An intuitive counterexample would be more easily generated once we look at its consequence of the necessity of distinctness

$$\Box \forall x \forall y (x \neq y \rightarrow \Box x \neq y) \quad \text{(ND)}$$

Consider: Some table, call it Woody, made out of material w is obviously distinct from some other table made out of material m . Yet, it is by no means necessary that Woody has to be made out of material w , and thus can be made out of material m (say m is some material similar to w). So, there is some possible world in which $x = y$, meaning $\Box x \neq y$ is false, and the principle of ND to be false. More easily, it seems plausible that there is some possible world in which water and H₂O are distinct, rendering NI false. So, if Sider’s premise 4 relies on NI and ND, then his argument runs into problems, for Williamson has no reason to accept NI.

I also suggest that while modal realism might apply to Williamson’s argument for accepting the Barcan Formula, it is not true that Williamson’s argument requires modal realism. The intuition runs like this: Lewis’s modal realism (about the way possible worlds are) is quite independent from the kinds of logic (and accessibility relations between worlds) we endorse. Lewis’s modal realism requires the principle of recombination because he intends to define possible worlds without the use of primitive modality and instead as a maximal spatiotemporally interrelated whole. But should Williamson instead endorse Stalnaker’s moderate realism or

some sort of ersatz realism, he does not require the principle of recombination that is the key piece of Sider's argument. It might very well be the case that Williamson is a modal realist, but it seems clear to me that talk about the systems of modal logic we should accept to describe modality runs quite independently from the ways in which possible worlds are. So, modal realism has little bearing on Williamson's many necessary existents, and Williamson can reject the principle of recombination in favour of a theory of possible worlds that accepts primitive modality (perhaps Rosen's fictionalism). In short, we can deny premise 3.

Therefore, Sider's argument does not really work – we can accept the iterative conception of the set, while committing to the mere existence of bare possibilia. We just have to deny certain auxiliary premises and modal realism.